INCH-POUND
MIL-DTL-55302/119H
w/AMENDMENT 3
18 October 2021
SUPERSEDING
MIL-DTL-55302/119H
w/AMENDMENT 2
19 September 2016

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

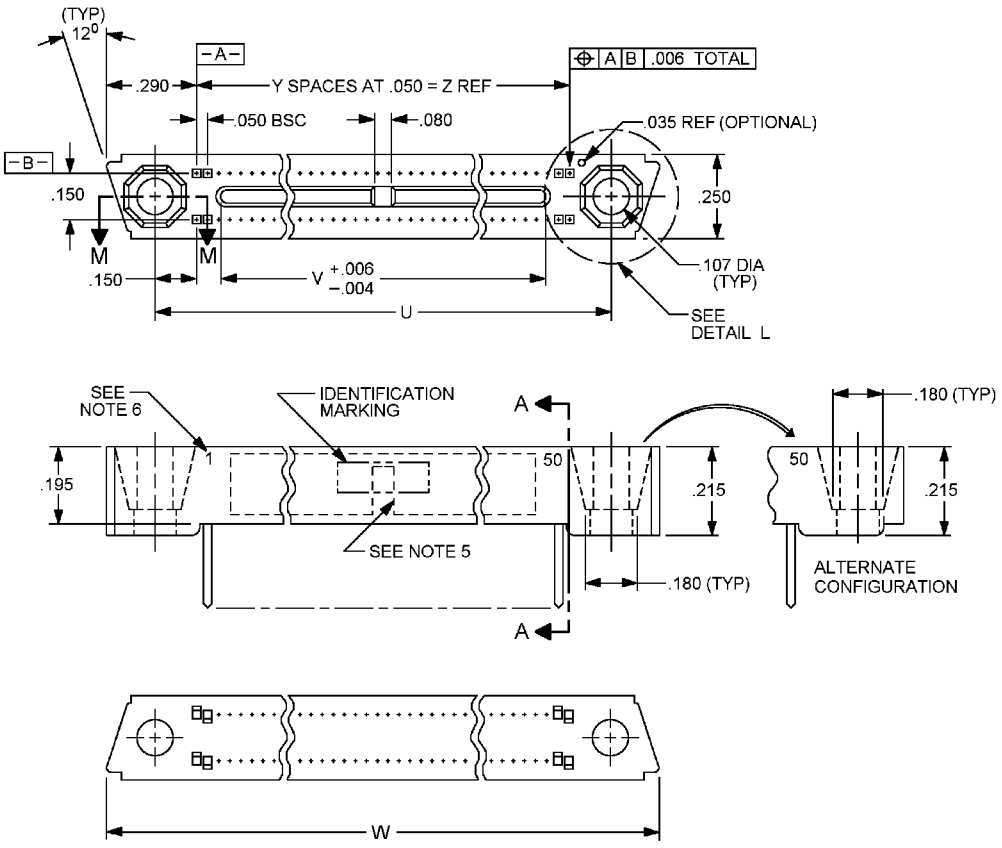
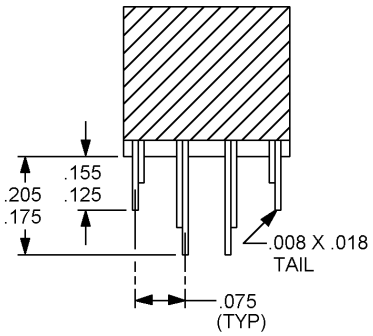


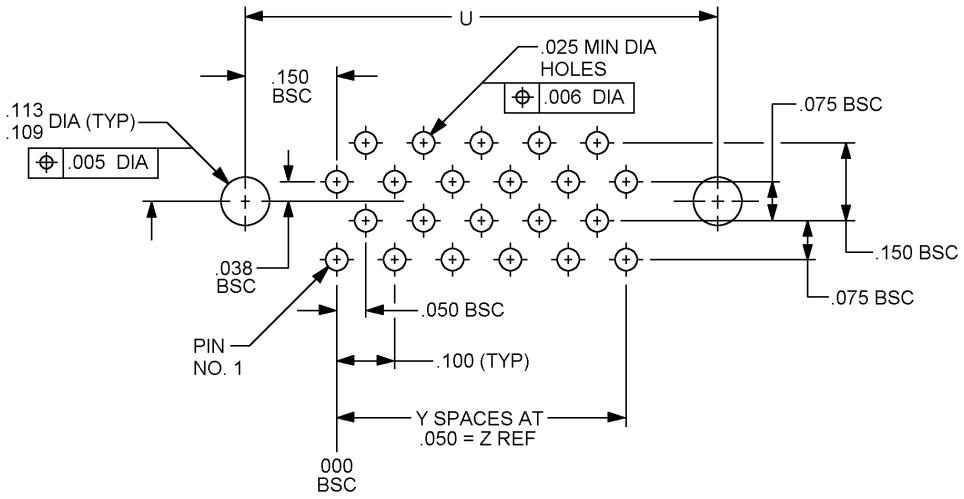
FIGURE 1. Connectors, receptacle, .050 (1.27 mm) spacing.

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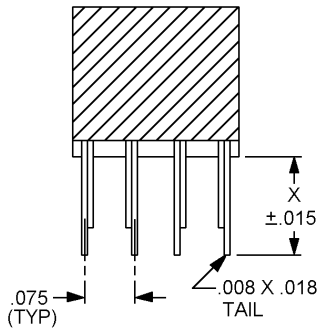
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SECTION A - A
TYPE I



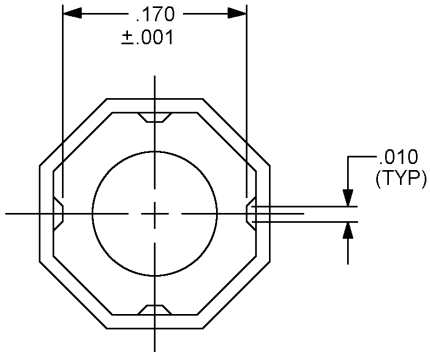
RECOMMENDED PRINTED WIRING BOARD
HOLE PATTERN, CONNECTOR SIDE



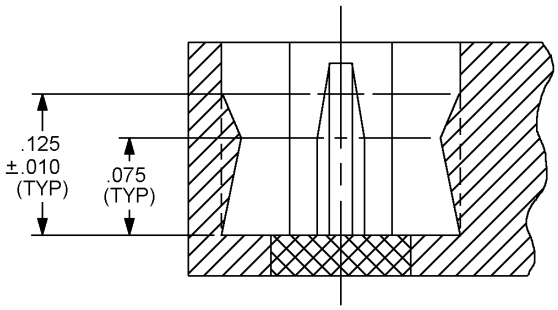
SECTION A - A
TYPE II

FIGURE 1. Connectors, receptacle, .050 (1.27 mm) spacing - Continued.

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DETAIL L



SECTION M - M

Inches	mm	Inches	mm	Inches	mm
.001	0.03	.050	1.27	.155	3.94
.004	0.10	.075	1.91	.170	4.32
.005	0.13	.080	2.03	.175	4.45
.006	0.15	.098	2.49	.180	4.57
.008	0.20	.100	2.54	.195	4.95
.010	0.25	.109	2.77	.205	5.21
.015	0.38	.087	2.21	.215	5.46
.018	0.46	.107	2.72	.250	6.35
.025	0.64	.113	2.87	.290	7.37
.035	0.89	.125	3.18		
.038	0.97	.150	3.81		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-DTL-55302/117 and MIL-DTL-55302/118.
5. Barrier is not present on 20 through 50 positions.
6. Numbers indicating end positions stamped this side only.

FIGURE 1. Connectors, receptacle, .050 (1.27 mm) spacing – Continued.

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TABLE I. Dash numbers and dimensions.

Type	Dash number	Number of contacts	Dimensions 1/						Contact Identification numbers
			U	V	W MAX	X	Y	Z	
I	01	20	.750 (19.50)	.300 (7.62)	1.040 (26.42)		9	.450 (11.43)	1-10
I	02	30	1.000 (25.40)	.550 (13.97)	1.290 (32.77)		14	.700 (17.78)	1-15
I	03	40	1.250 (31.75)	.800 (20.32)	1.540 (39.12)		19	.950 (24.13)	1-20
I	04	50	1.500 (38.10)	1.050 (26.67)	1.790 (45.47)		24	1.200 (30.48)	1-25
I	05	60	1.750 (44.45)	1.300 (33.02)	2.040 (51.82)		29	1.450 (36.83)	1-30
I	06	70	2.000 (50.80)	1.550 (39.37)	2.290 (58.17)		34	1.700 (43.18)	1-35
I	07	80	2.250 (57.15)	1.800 (45.72)	2.540 (64.52)		39	1.950 (49.53)	1-40
I	08	90	2.500 (63.50)	2.050 (52.07)	2.790 (70.87)		44	2.200 (55.88)	1-45
I	09	100	2.750 (69.85)	2.300 (58.42)	3.040 (77.22)		49	2.450 (62.23)	1-50
I	10	110	3.000 (76.20)	2.550 (64.77)	3.290 (83.57)		54	2.700 (68.58	1-55
I	11	128	3.450 (87.63)	3.000 (76.20)	3.740 (95.00)		63	3.150 (80.01)	1-64
II	12	100	2.750 (69.85)	2.300 (58.42)	3.040 (77.22)	.300 (7.62)	49	2.450 (62.23)	1-50
II	13	110	3.000 (76.20)	2.550 (64.77)	3.290 (83.57)	.300 (7.62)	54	2.700 (68.58)	1-55
II	14	128	3.450 (87.63)	3.000 (76.20)	3.740 (95.00)	.190 (4.83)	63	3.150 (80.01)	1-64

1/ Dimensions are in inches. Metric equivalents are given for information only.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Material:

Insulator body: Insulator material shall be in accordance with MIL-DTL-55302 or ASTM D5138 type LCP0120G30A43430 EA300ED035EE200EF140 or MIL-M-24519 type GLCP-30F.

Contact material: Contact material shall be in accordance with MIL-DTL-55302 or beryllium copper alloy C17400 in accordance with ASTM B768.

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Plating: Contact plating shall be gold over nickel in accordance with MIL-DTL-55302 on the contact engagement area for .100 inch (2.54 mm) minimum length. The termination area plating shall be tin lead over nickel in accordance with MIL-DTL-55302, to within .020 inches (0.51 mm) maximum of plastic housing body.

Contact identification: See figure 1.

Oversized pin exclusion: .0330 diameter pin.

Keying: See MIL-DTL-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .30 and the withdrawal force in pounds shall be a minimum of .03 times the number of contacts and shall not exceed the measured insertion force.

Contact engagement and separation force: Contact engagement and separation force shall be in accordance with MIL-DTL-55302 or EIA-364-37, test method C.

Engagement force: Maximum force to engage maximum test pin .0190 +.0000, -.0001, diameter shall be 6 ounces.

Separation force: Minimum force to extract minimum test pin .0170 +.0001, -.0000, diameter shall be .10 ounces.

Contact rating: 1.5 Amperes per contact, maximum.

Contact resistance: When mated with MIL-DTL-55302/117, the average contact resistance of all contacts measured shall not exceed .025 ohm, and no individual contact pair shall have a resistance exceeding .030 ohm. When mated with MIL-DTL-55302/118, the average contact resistance of all contacts measured shall not exceed 015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: Not applicable.

Operating temperature: -65°C to +125°C.

Dielectric withstanding voltage:

Sea level: 600 volts rms.

High altitude: 150 volts rms.

Solderability: Each connector shall be subjected to MIL-STD-202-208 (solder termination only) (applicable to qualification and retention of qualification only).

Part or Identifying Number (PIN): M55302/119 - (dash number from table 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item for the Government of for use in equipment to be delivered to the Government.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-55302, this document references the following:
MIL-DTL-55302/31
MIL-DTL-55302/117
MIL-DTL-55302/118
MIL-M-24519
MIL-STD-202-208
ASTM B768
ASTM D5138
EIA-364-37

CONCLUDING MATERIAL

Custodians:
Air Force - 85
DLA - CC

Review activity:
Air Force – 19

Preparing activity:
DLA - CC

(Project 5935-2021-118)

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